

STAT 339 METHODS OF LINEAR ALGEBRA

Linear Equations in Linear Algebra

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① LINEAR EQUATIONS IN LINEAR ALGEBRA

Systems of Linear Equations

Row Reduction and Echelon Forms

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Row Reduction and Echelon Forms

Systems of Linear Equations

Linear Equations

A **linear equation** in the variables is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

The coefficients $a_1, a_1, a_1, \dots, a_n$ are real numbers or complex numbers that are usually known in advance, and the constant term b is a real number. The number a_1 is the leading coefficient, and x_1 is the leading variable.

Systems of Linear Equations

A **system of linear equations** (or a **linear system**) is a collection of one or more linear equations involving the same variables - say $x_1, x_1, x_1, \dots, x_n$.

Systems of Linear Equations

Solution of a Linear Equation

A **solution** of the system is a list $(s_1, s_2, \dots, s_n)$ of numbers that makes each equation a true statement when the values $s_1, s_1, s_1, \dots, s_n$ are substituted for $x_1, x_1, x_1, \dots, x_n$, respectively.

Note

- The set of all possible solutions is called the **solution set** of the linear system,
- Two linear systems are called **equivalent** if they have the same solution set.

Systems of Linear Equations

Number of Solution of a System of Linear Equations

A system of linear equations has

- no solution, or
- exactly one solution, or
- infinitely many solutions

Systems of Linear Equations

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A system of linear equations has

- no solution, or
- exactly one solution, or
- infinitely many solutions

Note

- A system of linear equations is said to be **consistent** if it has either one solution or infinitely many solutions.
- A system of linear equation is said to be **inconsistent** if it has no solution.

Systems of Linear Equations

Matrix Notation

- The essential information of a linear system can be recorded compactly in a rectangular array called a matrix.
- For the following system of equations,

$$x_1 - 2x_2 + x_3 = 0$$

$$2x_2 - 8x_3 = 8$$

$$-4x_1 + 5x_2 + 9x_3 = -9$$

the matrix

Systems of Linear Equations

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the matrix

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix}_{3 \times 3}$$

is called the **coefficient matrix** of the system.

Systems of Linear Equations

Matrix Notation

- An **augmented matrix** of a system consists of the coefficient matrix with an added column containing the constants from the right sides of the equations.
- For the given system of equations in previous slide

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}_{3 \times 4}$$

is called the **augmented matrix** of the system.

Note

The basic strategy for solving a linear system is to replace one system with an equivalent system (i.e., one with the same solution set) that is easier to solve

Example (Solving System of Equations)

- Solve the given system of equations,

$$x_1 - 2x_2 + x_3 = 0$$

$$2x_2 - 8x_3 = 8$$

$$-4x_1 + 5x_2 + 9x_3 = -9$$

Systems of Linear Equations

Solution (Solving System of Equations)

- The elimination procedure is shown here with and without matrix notation, and the results are placed side by side for comparison*

$$\begin{aligned}x_1 - 2x_2 + x_3 &= 0 \\2x_2 - 8x_3 &= 8 \\-4x_1 + 5x_2 + 9x_3 &= -9\end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}$$

- Keep x_1 in the first equation and eliminate it from the other equations. To do so, add 4 times equation 1 to equation 3.*

Systems of Linear Equations

Solution (Solving System of Equations)

$$\begin{array}{r} 4x_1 - 8x_2 + 4x_3 = 0 \\ -4x_1 + 5x_2 + 9x_3 = -9 \\ \hline -3x_2 + 13x_3 = -9 \end{array}$$

- The result of this calculation is written in place of the original third equation*

$$\begin{array}{r} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -3x_2 + 13x_3 = -9 \end{array}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

Systems of Linear Equations

Solution (Solving System of Equations)

- Now, multiply equation 2 by $1/2$ in order to obtain 1 as the coefficient for x_2

$$x_1 - 2x_2 + x_3 = 0$$

$$x_2 - 4x_3 = 4$$

$$-3x_2 + 13x_3 = -9$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

- Use the x_2 in equation 2 to eliminate the $-3x_2$ in equation 3

$$3x_2 - 12x_3 = 12$$

$$-3x_2 + 13x_3 = -9$$

$$x_3 = 3$$

Systems of Linear Equations

Solution (Solving System of Equations)

- The new system has a triangular form*

$$x_1 - 2x_2 + x_3 = 0$$

$$x_2 - 4x_3 = 4$$

$$x_3 = 3$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

- Now, you want to eliminate the x_2 term from equation 1, but it is more efficient to use the x_3 term in equation 3 first to eliminate the $-4x_3$ and x_3 terms in equations 2 and 1.*

Systems of Linear Equations

Solution (Solving System of Equations)

- The new system has a triangular form*

$$\begin{array}{r} 4x_3 = 12 \\ 1x_2 - 4x_3 = 4 \\ \hline x_2 = 16 \end{array}$$

$$\begin{array}{r} -x_3 = -3 \\ x_1 - 2x_2 + x_3 = 0 \\ \hline x_1 - 2x_2 = -3 \end{array}$$

- Now, combine the results of these two operations*

Systems of Linear Equations

Solution (Solving System of Equations)

$$x_1 - 2x_2 = -3$$

$$x_2 = 16$$

$$x_3 = 3$$

$$\begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

- *Move back to the x_2 in equation 2, and use it to eliminate the $-2x_2$ above it. Because of the previous work with x_3 , there is now no arithmetic involving x_3 terms. Add 2 times equation 2 to equation 1 and obtain the system:*

Systems of Linear Equations

Solution (Solving System of Equations)

$$x_1 = 29$$

$$x_2 = 16$$

$$x_3 = 3$$

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

- Thus, the only solution of the original system is $(29, 16, 3)$. To verify that $(29, 16, 3)$ is a solution, substitute these values into the left side of the original system, and compute

Solution (Solving System of Equations)

$$(29) - 2(16) + (3) = 29 - 32 + 3 = 0$$

$$2(16) - 8(3) = 32 - 24 = 8$$

$$-4(29) + 15(16) + 9(3) = -116 + 80 + 27 = -9$$

- *The results agree with the right side of the original system, so $(29, 16, 3)$ is a solution of the system*

Elementary Row Operations

- Elementary row operations include the following:
 - ① (Replacement) Replace one row by the sum of itself and a multiple of another row
 - ② (Interchange) Interchange two rows.
 - ③ (Scaling) Multiply all entries in a row by a nonzero constant

Note

Two matrices are called **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other.

Elementary Row Operations

- It is important to note that row operations are reversible
- If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.
- Two fundamental questions about a linear system are as follows:
 - ① Is the system consistent; that is, does at least one solution exist?
 - ② If a solution exists, is it the only one; that is, is the solution unique?

Example (Existence and Uniqueness of System of Equations)

- Solve the given system of equations,

$$x_2 - 4x_3 = 8$$

$$2x_1 - 3x_2 + 2x_3 = 1$$

$$5x_1 - 8x_2 + 7x_3 = 1$$

Solution (Existence and Uniqueness of System of Equations)

- *The augmented matrix is*

$$x_2 - 4x_3 = 8$$

$$2x_1 - 3x_2 + 2x_3 = 1$$

$$5x_1 - 8x_2 + 7x_3 = 1$$

$$\begin{bmatrix} 0 & 1 & -4 & 8 \\ 2 & -3 & 2 & 1 \\ 5 & -8 & 7 & 1 \end{bmatrix}$$

- *To obtain an x_1 in in the first equation, interchange rows 1 and 2*

Solution (Existence and Uniqueness of System of Equations)

-

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 5 & -8 & 7 & 1 \end{bmatrix}$$

- *To eliminate the $5x_1$ term in the third equation, add $-5/2$ times row 1 to row 3.*

Solution (Existence and Uniqueness of System of Equations)

-

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & -1/2 & 2 & -3/2 \end{bmatrix}$$

- *Next, use the x_2 term in the second equation to eliminate the $-(1/2)x_2$ term from the third equation. Add $(1/2)$ times row 2 to row 3.*

Solution (Existence and Uniqueness of System of Equations)

- $$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

- The augmented matrix is now in triangular form. To interpret it correctly, go back to equation notation.*

Systems of Linear Equations

Solution (Existence and Uniqueness of System of Equations)

-

$$2x_1 - 3x_2 + 2x_3 = 1$$

$$x_1 - 4x_3 = 8$$

$$0 = 5/2$$

Systems of Linear Equations

Solution (Existence and Uniqueness of System of Equations)

- $$\begin{aligned}2x_1 - 3x_2 + 2x_3 &= 1 \\x_1 - 4x_3 &= 8 \\0 &= 5/2\end{aligned}$$

NOTES

- *The equation $0 = 5/2$ is a short form of $0x_1 + 0x_2 + 0x_3 = 5/2$*
- *There are no values of x_1, x_2, x_3 that satisfy last equation because the equation $0 = 5/2$ is never true.*
- *Since last equation and original equation have the same solution set, the original system is inconsistent (i.e., has no solution)*

Systems of Linear Equations

Exercise

- ① Solve in words the elementary row operation that should be performed on the system in order to solve it. [More than one answer is possible in 1(a).]

$$\begin{array}{rcll} (a) & x_1 + 4x_2 - 2x_3 + 8x_4 = 12 & (b) & x_1 - 3x_2 + 5x_3 - 2x_4 = 0 \\ & x_2 - 7x_3 + 2x_4 = -4 & & x_2 + 8x_3 = -4 \\ & 5x_3 - x_4 = 7 & & 2x_3 = 3 \\ & x_3 + 3x_4 = -5 & & x_4 = 1 \end{array}$$

- ② The augmented matrix of a linear system has been transformed by row operations into the form below. Determine if the system is consistent.

$$\left[\begin{array}{cccc} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array} \right]$$

Exercise

③ Is $(3, -4, 2)$ a solution to the following system?

$$5x_1 - x_2 + 2x_3 = 7$$

$$-2x_1 + 6x_2 + 9x_3 = 0$$

$$-7x_1 + 5x_2 - 3x_3 = -7$$

③ For what value of h and k is the following system is consistent?

$$2x_1 - x_2 = h$$

$$-6x_1 + 3x_2 = k$$

Definition

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:

- ① All nonzero rows are above any rows of all zeros.
- ② Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- ③ All entries in a column below a leading entry are zeros.

Reduced Echelon Forms

Definition

If a matrix in **echelon form** satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**):

- ① The leading entry in each nonzero row is 1.
- ② Each leading 1 is the only nonzero entry in its column

NOTES

- An **echelon matrix** (respectively, **reduced echelon matrix**) is one that is in **echelon form** (respectively, **reduced echelon form**.)

Reduced Echelon Forms

- Any nonzero matrix may be row reduced (i.e., transformed by elementary row operations) into more than one matrix in echelon form, using different sequences of row operations. However, the reduced echelon form one obtains from a matrix is unique.
- If a matrix A is row equivalent to an echelon matrix U , we call U an echelon form (or row echelon form) of A ; if U is in reduced echelon form, we call U the reduced echelon form of A .

Theorem (Theorem 1: Uniqueness of the Reduced Echelon Form)

Each matrix is row equivalent to one and only one reduced echelon matrix

Pivot Position

- A pivot position in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A .
- A pivot column is a column of A that contains a pivot position.

Example (1)

- Row reduce the matrix A below to echelon form, and locate the pivot columns of A.

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Row Reduction Algorithm

Solution

- *The top of the leftmost nonzero column is the first pivot position. A nonzero entry, or pivot, must be placed in this position.*
- *Now, interchange rows 1 and 4.*

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

- *Create zeros below the pivot, 1, by adding multiples of the first row to the rows below, and obtain the next matrix.*

Row Reduction Algorithm

Solution

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -15 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

- Choose 2 in the second row as the next pivot.
- Add $-5/2$ times row 2 to row 3, and add $3/2$ times row 2 to row 4.

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \end{bmatrix}$$

Row Reduction Algorithm

Solution

- *There is no way a leading entry can be created in column 3. But, if we interchange rows 3 and 4, we can produce a leading entry in column 4.*

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- *The matrix is in echelon form and thus reveals that columns 1, 2, and 4 of A are pivot columns.*
- *The pivots in the example are 1, 2 and -5*

Example (2)

- Apply elementary row operations to transform the following matrix first into echelon form and then into reduced echelon form.

$$\begin{bmatrix} 0 & -3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Row Reduction Algorithm

Solution

- **STEP 1:** *Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.*

$$\begin{bmatrix} 0 & -3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

- **STEP 2:** *Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.*

Row Reduction Algorithm

Solution

- *Interchange rows 1 and 3. (Rows 1 and 2 could have also been interchanged instead.)*

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & -3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

- **STEP 3:** *Use row replacement operations to create zeros in all positions below the pivot.*

Row Reduction Algorithm

Solution

- *We could have divided the top row by the pivot, 3, but with two 3s in column 1, it is just as easy to add times row 1 to row 2.*

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

- **STEP 4:** *Cover the row containing the pivot position, and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify*

Row Reduction Algorithm

Solution

- *With row 1 covered, step 1 shows that column 2 is the next pivot column; for step 2, select as a pivot the “top” entry in that column.*

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

- *For step 3, we could insert an optional step of dividing the “top” row of the submatrix by the pivot, 2. Instead, we add $-3/2$ times the “top” row to the row below.*

Example

- This produces the following matrix.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- When we cover the row containing the second pivot position for step 4, we are left with a new submatrix that has only one row.

Solution

-

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- *Steps 1–3 require no work for this submatrix, and we have reached an echelon form of the full matrix. We perform one more step to obtain the reduced echelon form.*
- **STEP 5:** *Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.*

Row Reduction Algorithm

Solution

- The rightmost pivot is in row 3. Create zeros above it, adding suitable multiples of row 3 to rows 2 and 1.

① add -6 times row 3 to row 1 $\Rightarrow R1$

② add -2 times row 3 to row 2 $\Rightarrow R2$

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- The next pivot is in row 2. Scale this row, dividing by the pivot.

① $1/2$ times row 2 $\Rightarrow R2$

Row Reduction Algorithm

Solution

-

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- *Create a zero in column 2 by adding 9 times row 2 to row 1.*

Row Reduction Algorithm

Solution

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$$\begin{bmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- *Finally, scale row 1, dividing by the pivot, 3.*

Row Reduction Algorithm

Solution

-

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- *This is the reduced echelon form of the original matrix.*
- *The combination of steps 1–4 is called the forward phase of the row reduction algorithm. Step 5, which produces the unique reduced echelon form, is called the backward phase.*

Solutions of Linear Systems

- The row reduction algorithm leads to an explicit description of the solution set of a linear system when the algorithm is applied to the augmented matrix of the system.
- Suppose that the augmented matrix of a linear system has been changed into the equivalent reduced echelon form.

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are 3 variables because the augmented matrix has four columns. The associated system of equations is

Solutions of Linear Systems



$$x_1 - 5x_3 = 1$$

$$x_2 + x_3 = 4$$

$$0 = 0$$

(1)//

- The variables x_1 and x_2 corresponding to pivot columns in the matrix are called basic variables. The other variable, x_3 , is called a free variable.
- Whenever a system is consistent, as in (1), the solution set can be described explicitly by solving the reduced system of equations for the basic variables in terms of the free variables.
- This operation is possible because the reduced echelon form places each basic variable in one and only one equation.

Solutions of Linear Systems

- In (1), solve the first and second equations for x_1 and x_2 . (Ignore the third equation; it offers no restriction on the variables.)

$$x_1 = 1 + 5x_3$$

$$x_2 = 4 - x_3$$

x_3 is free

(2)//

- The statement " x_3 free" means that you are free to choose any value for x_3 . Once that is done, the formulas in (2) determine the values for x_1 and x_2 . For instance, when $x_3 = 0$, the solution is (1,4,0); when $x_3 = 1$, the solution is (6,3,1).
- Each different choice of x_3 determines a (different) solution of the system, and every solution of the system is determined by a choice of x_3

Parametric Description of Solution Sets

- The description in (2) is a parametric description of solutions sets in which the free variables act as parameters.
- Solving a system amounts to finding a parametric description of the solution set or determining that the solution set is empty.
- Whenever a system is consistent and has free variables, the solution set has many parametric descriptions.

Parametric Description of Solution Sets

- For instance, in system (1), add 5 times equation 2 to equation 1 and obtain the following equivalent system.

$$x_1 + 5x_2 = 21$$

$$x_2 + x_3 = 4$$

- We could treat x_2 as a parameter and solve for x_1 and x_3 in terms of x_2 , and we would have an accurate description of the solution set.
- When a system is inconsistent, the solution set is empty, even when the system has free variables. In this case, the solution set has no parametric representation.

Existence and Uniqueness theorem

Theorem (Theorem 2: Existence and Uniqueness Theorem)

A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column—i.e., if and only if an echelon form of the augmented matrix has no row of the form

$$[0 \cdots 0 \ b]$$

with b nonzero.

- If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

Row Reduction to Solve a Linear System

- Write the augmented matrix of the system.
- Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
- Continue row reduction to obtain the reduced echelon form.
- Write the system of equations corresponding to the matrix obtained in step 3.
- Rewrite each nonzero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

Row Reduction and Echelon Forms

Exercise

- ① Find the general solution to the linear system whose augmented matrix is

$$\begin{bmatrix} 1 & -3 & -5 & 0 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

- ② Find the general solution to the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= 0 \\ -2x_1 + 4x_2 + 5x_3 - 5x_4 &= 3 \\ 3x_1 - 6x_2 - 6x_3 + 8x_4 &= 2 \end{aligned}$$